System Resilience: Technical Worksheet

Grade 12 · Worksheet · About 25 minutes
Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

How to use this worksheet

- Do the parts **in order** from top to bottom.
- Each part has a **time** so you know how long to spend.
- Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: System resilience is the ability of a system to keep delivering essential functions during stress—and to recover quickly after disruption. You will connect resilience to **redundancy**, **backups**, **failover**, monitoring, and recovery objectives (RTO/RPO concepts at a high level).

Do this:

1. **In one sentence**, what does resilience mean beyond “never fails”?

   _____

   _____

2. **Name one system** (school, city, or personal) that must keep working under stress.

   _____

Part 2 – Resilience vs mere uptime (about 5 minutes)

Uptime measures how often a system is available. **Resilience** asks what happens when parts fail: Can users still complete critical tasks? How fast can the system return to a safe state? A system can have high past uptime and still be fragile if one failure cascades.

> **Real-life scenario:** A school stores all attendance, grades, and lesson materials on a single server with no tested backup. The server has been "up" for 18 months. One disk failure takes everything offline for a week.

Discuss with a partner or reflect in writing: Why was long uptime not the same as resilience here?

Do this: Label each property as supporting **Uptime focus** or **Resilience focus** (best fit).

Property	Focus
Percentage of hours the portal was reachable last year	_____
Documented failover to a secondary site within 2 hours	_____
Regular restore tests from backup	_____
"We have never had an outage, so we skip drills"	_____

Part 3 – Redundancy, backups, and failover (about 5 minutes)

Redundancy means extra capacity or alternate paths (second power supply, spare Wi-Fi AP, mirrored disk). A **backup** is a copy of data stored separately. **Failover** is switching to a standby when the primary fails. Backups without restore tests are optimism, not resilience.

> **Real-life scenario:** IT keeps nightly backups on the same rack as the production server. A room fire destroys both. Another school keeps offline and off-site copies and practices restoring a sample folder each quarter.

Critical thinking: Which school is more resilient, and which single design choice mattered most?

Do this: Match each control to its primary role.

Word bank: *Redundancy · Backup · Failover · Monitoring*

Control	Role
Second internet link that activates if the first dies	_____
Encrypted copy of databases stored off-site	_____
Alerts when disk errors rise above a threshold	_____
Automatic switch from primary app server to standby	_____

Part 4 – Recovery objectives and single points of failure (about 5 minutes)

A **single point of failure (SPOF)** is one component whose loss stops the whole service. **RTO** (recovery time objective) is how quickly you need to be back; **RPO** (recovery point objective) is how much data loss you can tolerate (e.g. last 15 minutes). Clear objectives guide design choices.

> **Real-life scenario:** An online exam platform can lose at most 5 minutes of answers (tight RPO) and must reopen within 30 minutes (RTO). Designers choose frequent replication plus a hot standby—not weekly tape-only backups.

Do this:

1. **Circle** the SPOF in this setup: One authentication server · Two mirrored databases · Dual power feeds
2. **For a school grading portal the night before report cards**, propose a rough RTO and RPO in plain language (not exact numbers required—reasonable ranges).
   RTO idea:

   RPO idea:

   ---
3. **Explain** one trade-off: stronger resilience often costs more money or complexity. Give one example.

Part 5 – Resilience architecture sketch (about 5 minutes)

Do this:

1. **Sketch a resilient mini-architecture** for one critical school service (portal, Wi-Fi, or file storage). Include at least: primary, backup/redundancy, monitoring, and a recovery note.

2. **Star the SPOF** you removed (or still worry about) and say why.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

☐ I can distinguish **uptime** from **resilience**.

☐ I can explain **redundancy**, **backup**, **failover**, and **monitoring**.

☐ I can identify a **single point of failure**.

☐ I can discuss **RTO/RPO** ideas in plain language.

☐ I can sketch a simple resilient design with recovery in mind.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on System Resilience (Technical). Central topic: System resilience. Main technology area: Reliability engineering / continuity.